

# Daheng Yin

dya64@sfu.ca | Simon Fraser University, Burnaby, Canada

## Education

### Simon Fraser Univ.

Ph.D. in Computing Science  
Advisor: Prof. Jiangchuan Liu  
2023.08 -- Present

### Southeast Univ.

M.Eng. in Computer Science  
Advisor: Prof. Fang Dong  
2020.08 -- 2023.07

### Jiangnan Univ.

B.Eng. in IoT Engineering  
Advisor: Prof. Ya Guo  
2016.08 -- 2020.07  
GPA: 3.63, Rank 9/141

## Research Areas

- Volumetric Video Streaming
- Dynamic 3D Reconstruction
- Embodied AI Systems

## Links

Homepage:  
[yindaheng98.top](http://yindaheng98.top)  
Github: [@yindaheng98](https://github.com/yindaheng98)

## Research Profile

- I build real-time 4D spatial perception infrastructure for Physical AI: systems that re-construct, compress, and stream photorealistic 4D spatial representations of dynamic physical environments, enabling Physical AI systems to perceive the physical world, interact with it, and access it remotely.

## Selected Publications

### CAGS: Color-Adaptive Volumetric Video Streaming with Dynamic 3D

#### Gaussian Splatting SIGGRAPH, 2026

- **Daheng Yin**, Yili Jin, Jianxin Shi, Isaac Ding, Miao Zhang, Fangxin Wang, Zhaowu Huang, Cong Zhang, Jiangchuan Liu, Fang Dong
- We propose color-adaptive 3DGS streaming for remote physical environments: compresses dynamic 3DGS volumetric video into vector-quantized LoD representations and restores color fidelity on the client, preserving photorealistic access under fluctuating bandwidth.

### TrackerSplat: Exploiting Point Tracking for Fast and Robust Dynamic 3D

#### Gaussians Reconstruction SIGGRAPH Asia, 2025

- **Daheng Yin**, Isaac Ding, Yili Jin, Jianxin Shi, Jiangchuan Liu
- High-performance dynamic 3D scene reconstruction for fast-moving scenes with point tracking: lifts cross-frame point trajectories onto 3D Gaussians and guides their translation, rotation, and scale compensation, reducing drift, artifacts, and temporal flickering.

### FSVFG: Towards Immersive Full-Scene Volumetric Video Streaming with

#### Adaptive Feature Grid ACM Multimedia, 2024

- **Daheng Yin**, Jianxin Shi, Miao Zhang, Zhaowu Huang, Jiangchuan Liu, Fang Dong
- Adaptive feature-grid streaming for large-scale full-scene volumetric video: represents wide-area dynamic scenes with compact feature grids and selectively transmits features and residuals, balancing scene coverage, visual quality, and transmission cost.

## Teaching Experiences

### CMPT 310 Introduction to Artificial Intelligence Summer 2026

- Topics: Search Algorithms, Supervised Learning, Gradient Descent, Markov Decision Processes, Bayesian Networks.

### CMPT 365 Multimedia Systems Fall 2024

- Topics: Multimedia Representation, Audio/Video Compression, Image and Video Processing, Multimedia Communication.

### CMPT 300 Operating Systems I Fall 2023

- Topics: Process Management, Synchronization, Memory Management, File Systems, Multiprogramming.

## Contests & Honors

|      |                                                   |                        |
|------|---------------------------------------------------|------------------------|
| 2025 | SIGGRAPH Asia Volumetric Video Workshop           | Compression Track 3rd  |
| 2022 | TensorRT Hackathon (NVIDIA & TIANCHI)             | Winner Prize           |
| 2021 | TensorRT Hackathon (NVIDIA & TIANCHI)             | Rank 4/48              |
| 2020 | Outstanding Graduate, Jiangnan University         |                        |
| 2018 | National College Mathematical Contest in Modeling | 2nd Prize (National)   |
| 2017 | China National Scholarship (2016-2017)            |                        |
| 2017 | 9th National College Mathematical Contest         | 2nd Prize (Provincial) |
| 2017 | 14th Jiangsu College Mathematical Contest         | 1st Prize              |